

Name: _____

Score: _____ / _____

Quiz 1

Part 1: Quiz 1

1

Σ^* always contains the empty string as an element, when Σ is an alphabet.

- A.
True
- B.
False
- C.
None of the above

Answer Point Value: 1.0 points

Answer Key: A

2

Let set A have c elements and set B have d elements. How many elements are in

(a) $A \times B$

(b) Powerset of A?

A.

(a) c times d

(b) 2^c

B.

(a) c^d

(b) c times d

C.

(a) c times d

(b) c^c

D.

(a) c times d

(b) d^c

Answer Point Value: 1.0 points

Answer Key: A

3

List one string in L and one string not in L, for

$L = \{ w \text{ is in } \{a\}^* \mid \text{there is some string } u \text{ such that } ww = uu \}$

A.

String in L is empty string

String not in L is aa

B.

String not in L is empty string

String in L is aa

C.

String in L is aaa

String not in L is empty string

D.

None of the above

Answer Point Value: 1.0 points

Answer Key: D

4

Let $L_1 = \{\emptyset\}$ and $L_2 = \{a\}$. Describe precisely the strings in

$$L_3 = L_1 \cdot (L_2^*)$$

$$L_4 = L_2 \cdot (L_1^*)$$

A.

$$L_3 = \{a\}$$

$$L_4 = \{\emptyset\}$$

B.

$$L_3 = \{\emptyset\}$$

$$L_4 = \{a\}$$

C.

$$L_3 = \{a^*\}$$

$$L_4 = \{\emptyset\}$$

D.

$$L_3 = \{\emptyset\}$$

$$L_4 = \{a^*\}$$

Answer Point Value: 1.0 points

Answer Key: B

5

It is always possible to modify a given DFA D to D' such that D' still accepts the same language as D , but D' has only one accepting state.

A. False

B.

True

Answer Point Value: 1.0 points

Answer Key: A

6

It is always possible to modify a given DFA D to D' such that D' still accepts the same language as D , but the number of states in D' are prime.

A.

True

B.

False

Answer Point Value: 1.0 points

Answer Key: A

7

All regular languages are finite.

- A. True
- B. False

Answer Point Value: 1.0 points
Answer Key: B

8

Any subset of a regular language is always a regular language.

- A. True
- B. False

Answer Point Value: 1.0 points
Answer Key: B

9

There exists a finite language that is not regular.

- True
- False

Answer Point Value: 1.0 points
Answer Key: False

10

There is a Non-Deterministic Finite Automata that recognizes the language $L = \{a^{n^2} \mid 0 \leq n \leq 100\}$.

- A.
False
- B.
True

Answer Point Value: 1.0 points
Answer Key: B

11

If L is an infinite language and L is accepted by a Deterministic Finite Automata, then the complement of L is a finite language.

True

False

Answer Point Value: 1.0 points

Answer Key: False

12

Let $L = \{ w \text{ is in } \{a,b\}^* \mid w \text{ has equal number of a's and b's} \}$.
 Let $L1 = L \text{ intersect } a^* b^*$.

Choose the correct characterization of $L1$.

A.

$$L1 = \{a^* b^*\}$$

B.

$$L1 = \{a^j b^k \mid j, k \geq 0\}$$

C.

$$L1 = \{a^j b^k \mid j \text{ is not equal to } k\}$$

D.

None of the above

Answer Point Value: 1.0 points

Answer Key: D